

Problem

A 4-kVA, 2,400/240-V rms transformer has 250 turns on the primary side. Calculate:

(a) the turns ratio,

(b) the number of turns on the secondary side,

(c) the primary and secondary currents.

Step-by-step solution

Step 1 of 3

$$\begin{aligned} \text{(A) } n &= \frac{V_2}{V_1} \\ &= \frac{240}{2400} = 0.1 . \end{aligned}$$

Step 2 of 3

$$\begin{aligned} \text{(B) } n &= \frac{N_2}{N_1} \\ \text{Or,} \\ N_2 &= n N_1 \\ &= 0.1 (250) = 25 \text{ Turns} . \end{aligned}$$

Step 3 of 3

$$\begin{aligned} \text{(C) } S &= I_1 V_1 \\ \text{Or,} \\ I_1 &= \frac{S}{V_1} \\ &= \frac{4 \times 10^3}{2400} = 1.6667 \text{ A} \\ S &= I_2 V_2 \\ \text{Or,} \\ I_2 &= \frac{S}{V_2} \\ &= \frac{4 \times 10^4}{240} = 16.667 \text{ A} . \end{aligned}$$